

SAMPLED ANSWERS

5 samples, 4 unique after dedup

William Golding

Golding

The author is William Golding

Peter Benchley

for the question "Who wrote Lord of the Flies?"

BIDIRECTIONAL ENTAILMENT

4 answers, 6 pairs, 12 DeBERTa forward inputs

one pair, run both ways

Q + a1 [SEP] Q + a2 → entailment

Q + a2 [SEP] Q + a1 → neutral

neither direction returned contradiction

so a2 is merged into a1's semantic set

a single contradiction, either way, keeps the two apart

argmax over the 3-way MNLI head; only label 0 (contradiction) splits a pair

SEMANTIC SETS

clusters, not strings, are what get counted

SET 1

William Golding

Golding

The author is William Golding

SET 2

Peter Benchley

number_of_semantic_sets = 2

PREDICTIVE ENTROPY — OVER SEQUENCES

$$PE = -(1/N) * \sum_i \log p(s_i | x)$$

Every sample is its own outcome. Three ways of saying Golding count as three, so a model that is confident but verbose reads as uncertain.

length-normalised variant: average_predictive_entropy

SEMANTIC ENTROPY — OVER MEANING SETS

$$SE = -(1/|C|) * \sum_c \log \text{sumexp}_{i \in c} \log p(s_i | x)$$

Likelihood is pooled inside each set before the average, so the three Golding samples contribute one term. Only real disagreement survives.

code subtracts --llh_shift (default 5.0) from each pooled term